SOMAIYA VIDYAVIHAR UNIVERSITY

15.05.2025(M)

SET - B

Semester: January 2025– April 2025		
Maximum Marks: 50 Examination: End-Semester Examination Duration: 2 Hrs. 30 min		
Programme code: H05 Programme: Honors Software and Mobile Application Development	Class: S.Y.	Semester: IV (SVU 2023)
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department: COMPUTER	
Course Code: 216H05C401	Name of the Course: Fundamental of Swift	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any **four**	20
a	Explain error handling in Swift? Describe the do-try-catch block with example.	05
b	Explain type constraints in generics? Give an example.	05
c	How do you access, add, and update values in a dictionary? Explain with example.	05
d	Compare and contrast the for-in, while, and repeat-while loops in Swift. Give examples to explain how and when to use each.	05
e	Explain substrings in Swift? Write code to extract a substring from a given string	05
f	Explain the significance of protocol oriented programing and how it is different from object oriented programming	05
Q.2	Attempt any **One**	10
a	Write a swift program to perform following • Create a User struct with username, password, and optional email. • Write a login(user:) function that uses guard to validate non-empty username and password, and exits with an error if validation fails. • If valid, print a welcome message and unwrap email using if let to display if available. • Demonstrate the function with valid and invalid User instances.	
b	Write a swift program to perform following for accessing User Info: • Create a struct named UserProfile with a name (String) and an optional age (Int?). • Write a function that takes a UserProfile instance. • Use optional binding (if let) to safely unwrap the age property. • If age has a value, print "User is [age] years old." Otherwise, print, "Age is not available." • Demonstrate creating a UserProfile with and without an age and calling the function.	
Q.3	Attempt the following	20
	Using UIkit in StoryBoard, Create a login screen with the following fields: • Username (UITextField) • Password (UITextField)	

	• Login (UIButton) • Reset (UIButton) All arranged in a **Vertical StackView** with spacing. • A UISwitch to indicate dark mode preference. **Requirements:** • Add Appropriate Auto Layout constraints to make the UI responsive across all screen sizes. • To display the entered user details using UILabel, add a second screen and execute a navigation segue upon submission. • Implement logic in Reset to clear both text fields.	